

Manufacturing Anomaly Detection, Root-Cause Analytics & CAPA Decision System

Abstract

This study develops and stress-tests a synthetic manufacturing quality analytics architecture over **504,000 production events**, **18,000 vehicles**, **28 stations**, **56 machines**, and **14 heterogeneous injected failure mechanisms**. The central finding is methodological rather than algorithmic: apparent anomaly performance and intuitive Pareto root-cause narratives weaken materially when temporal leakage, fixed investigation capacity, confounding, censoring, placebo tests, and economic uncertainty are treated explicitly.

At a fixed 0.25% review budget, the strongest benchmark detector by lift is **OneClassSVM**, with precision **13.01%**, recall **1.60%**, and lift **8.20×** over random review. Absolute recall remains low, demonstrating that a narrow quality detector cannot substitute for dedicated downtime/reliability monitoring. A random supervised split inflates PR-AUC by **63.5%** versus chronological evaluation.

Known-truth RCA Monte Carlo yields **96.00% Top-1 recovery**, **0.980 MRR**, **97.50% 95% interval coverage**, and **4.50% false RCA declarations under the null**. The focal ST07 CAPA has a negative DiD estimate (-0.0025), but non-flat pre-trends and coarse placebo distributions downgrade causal evidence to **Inconclusive**. Probabilistic economics remain decision-relevant but uncertain: median 5-year NPV is **€62,325**, $P(\text{NPV} > 0)$ is **72.76%**, while 5% CVaR is **€-150,116**.

1. Research questions

RQ1. How much do temporal validation and fixed review budgets change the apparent performance of anomaly detection?

RQ2. How materially can confounding alter manufacturing root-cause rankings?

RQ3. Can a known-truth synthetic experiment quantify root-cause recovery and false-cause discovery?

RQ4. How robust is CAPA effect estimation with one treated station and a limited donor pool?

RQ5. How does causal uncertainty propagate into a CAPA investment decision?

2. Industrial and methodological context

The architecture is designed around the distinction between monitoring, diagnosis, causal attribution, corrective action, and economics. These are different inference tasks. An anomaly indicates deviation; it does not identify responsibility. A predictive association can prioritize investigation; it does not prove

mechanism. A before/after decline can motivate verification; it does not by itself identify a treatment effect.

The 2026 AIAG & VDA SPC framework emphasizes a control-loop view linking process knowledge, statistical analysis, production readiness and Control Plan execution. The repository therefore terminates not at a dashboard, but at reaction plans, evidence grades, and a decision model. Measurement and reliability components follow the same discipline: measurement error is separated from latent process truth, and repairable-system reliability is separated from censored lifetime estimation.

3. Synthetic data-generating process

The DGP spans 300 days with 60 vehicles/day and one record per vehicle-station visit. Four vehicle variants, three shifts, supplier lots, machine assignment, operators, maintenance exposure, environmental variation, measurement error, missingness and label imperfections generate overlapping sources of variation. Fourteen failure mechanisms cover gradual drift, step change, variance inflation, short intense failures, lot excursions, machine wear, interaction effects, recurrent sensor anomalies and temporary issues.

A deliberate operator-machine confounding pattern is introduced at ST07. OP-07-01 is disproportionately assigned to M07B while receiving a modest conditional protective effect. This permits the study to test whether raw operator ranking survives adjustment. The final adjusted operator odds ratio is 0.77, with a wide interval, illustrating that the aggregate signal is not sufficient evidence of operator responsibility.

4. Staged reproducibility architecture

The release replaces a monolithic pipeline with ten independent stages: generation, validation, feature governance, SPC, anomaly detection, RCA, reliability, causal CAPA, economics and reporting. Each stage fingerprints its configuration and inputs, writes intermediate artifacts, records runtime/peak memory, and can be resumed independently. This architecture is part of the methodological result: reproducibility requires that validation experiments be runnable without reloading every upstream DataFrame.

5. Data quality and contracts

Five machine-readable contracts cover production events, telemetry, maintenance, supplier lots and CAPA. They enforce key uniqueness, non-negative durations, category validity and time ordering. The final data-quality audit contains no blocking contract errors. Synthetic missingness and sensor uncertainty are retained by design rather than cleaned away as if they were mistakes.

6. Feature governance and ablation

The initial detector candidate set included cycle-time, process, environment, tool-age and changeover information. Multi-seed ablation falsified the assumption that more industrially plausible features necessarily improve the quality detector. Removing environment, tool-age and cycle-time each improved PR-AUC in all six ablation seeds; removing process signal degraded it in all six. The final early-warning quality detector therefore uses the process signal as its primary feature, while cycle, tool-age and environment remain available to diagnostic/reliability layers.

7. Rare-event anomaly evaluation

ROC-AUC is secondary because anomalies are rare. The primary operating point is a review capacity of 0.25% of evaluated events. Under the final temporal benchmark, OneClassSVM reviews only that narrow tail and yields lift $8.20\times$. This is useful prioritization but not broad anomaly coverage: recall is only 1.60%. The correct operational interpretation is therefore **queue prioritization**, not autonomous defect detection.

The investigation economics table converts 0.10%, 0.25%, 0.50% and 1.00% review budgets into events, true anomalies, false investigations, analyst hours and cost. This exposes alert fatigue directly.

8. Temporal leakage experiment

The supervised probe is deliberately not the production detector. It exists to measure split sensitivity. Chronological PR-AUC is 0.0322; random-split PR-AUC is 0.0527, an inflation of 63.5%. This result validates the decision to prohibit random split as the sole performance estimate.

9. SPC policy evidence

EWMA and CUSUM Monte Carlo experiments distinguish in-control ARL from step, drift and variance-change behavior. Estimated in-control mean run lengths are approximately 568 (EWMA) and 473 (CUSUM). For a small 0.5-sigma step they fall to roughly 42 and 37 observations. These are policy-comparison simulations, not exact plant ARLs.

10. Failure-regime capability

The failure $\times$ detector matrix makes a crucial limitation visible: process-value detectors detect some drift/variance mechanisms but often miss downtime-centric or interaction failures. This is why the final architecture does not collapse SPC, condition monitoring and quality anomaly ranking into one ensemble score. Specialized monitors remain necessary.

11. Root-cause analysis

RCA proceeds from raw screening to adjusted evidence. Standard logistic, Firth penalized likelihood and a transparent hierarchical Beta-Binomial comparator are used around the ST07 machine contrast. M07B odds ratios are approximately 2.58 (standard) and 2.49 (Firth). Agreement strengthens the association, but physical causality is not inferred from regression alone.

The deliberate confounding case changes interpretation after adjustment. Raw operator-level rates are affected by assignment to M07B; the adjusted operator effect is not statistically decisive. This result directly falsifies operator blame based on raw Pareto.

12. Known-truth Monte Carlo validation

Across 200 synthetic replications with a known true machine effect, Top-1 RCA recovery is 96.00%, MRR is 0.980, effect RMSE is 0.338 log-odds and 95% CI coverage is 97.50%. Under the null, false supported RCA declarations occur in 4.50% of replications and the system returns inconclusive in 95.50%. This ability to abstain is treated as a feature.

13. Reliability

Machine reliability reporting uses MTBF/MTTR/availability as descriptive repairable-system summaries, right-censored Weibull likelihood for interval-life diagnostics, and Crow-AMSAA separately for recurrent-failure trend. The repository explicitly rejects the use of a failure-only Weibull fit as a universal machine-reliability model.

14. CAPA causal evidence

The focal ST07 DiD estimate is -0.00248 with CI [-0.00384, -0.00112]. Taken alone, this appears compelling. It is not accepted alone. The event study contains materially non-zero pre-period effects, synthetic-control evidence has limited donor support, placebo-time empirical p is 0.600, and placebo-space empirical p is 0.364. Final CAPA evidence is therefore **Inconclusive**, not Strong.

Subgroup analysis is performed at station-day-subgroup level for shift and vehicle model. Machine subgroup results are labelled descriptive because a within-ST07 machine comparison lacks the same cross-station counterfactual.

15. Economic decision analysis

The financial model explicitly propagates uncertainty rather than multiplying a before/after difference by production volume. Because causal diagnostics are weak, the DiD absolute effect is multiplied by a credibility factor of 0.45. The resulting NPV distribution has P5 €-97,369, P50 €62,325, P95 €318,261,

$P(NPV > 0)$ 72.76%, and 5% CVaR €-150,116. The tail therefore contains economically negative outcomes even though the median is positive.

16. Claim governance and falsification

Every major claim is paired with evidence, an alternative specification, a targeted falsification test and a status. This produces several negative results: the full feature set is rejected; raw operator blame is rejected; a strong ST07 causal claim is rejected; deterministic positive NPV is rejected. These failures increase credibility because the analysis is permitted to change its conclusion.

17. Control and reaction plan

Five failure types are mapped to signals, thresholds, owners, containment, diagnostic actions, escalation and verification. This links analytics to operational response rather than treating model output as the terminal deliverable.

18. Cost of Quality

The synthetic decision layer distinguishes prevention, appraisal, internal failure and external failure. External failure value is intentionally excluded from the base case when it is not directly represented by the DGP, avoiding fabricated warranty savings.

19. Runtime and scalability

The runtime tables report stage-level peak RSS plus feature-engineering scaling at 100k, 250k and 504k events. The primary release target is reproducibility on a workstation, not real-time streaming latency. Intermediate compression uses gzip-pickle in the verified environment because `pyarrow` was unavailable; Parquet is the recommended deployment representation.

20. Limitations and external validity

This is a synthetic study. It does not validate an OEM process, Ferrari production system, operator performance, supplier quality or financial case. External deployment would require real MES histories, QMS nonconformity records, CMMS work orders, supplier genealogy, measurement-system studies, operator assignment histories, CAPA documentation and finance-approved cost data. The current work demonstrates methodological discipline and integration, not external empirical truth.

21. Conclusion

The strongest contribution is the sequence of falsifications. Raw anomaly performance looked stronger before leakage controls. More features looked better

before multi-seed ablation. Raw operator rates looked actionable before machine adjustment. DiD looked causal before event-study and placebo diagnostics. Median NPV looked decisive before downside simulation. The final repository is therefore smaller in claims but stronger in credibility.

References / methodological anchors

- NIST/SEMATECH e-Handbook of Statistical Methods: SPC, process capability, measurement systems, censored reliability and repairable systems.
- AIAG & VDA SPC Manual, 1st ed. (2026), and AIAG Control Plan, 1st ed. (2024).
- ASQ Cost of Quality framework (prevention, appraisal, internal failure, external failure).
- Firth penalized-likelihood literature for sparse/separated logistic regression.
- Journal of Manufacturing Systems literature on robustness and causal challenges in manufacturing RCA.

Appendix A - Primary detector benchmark

[illegible]

Appendix B - RCA recovery summary

replication	top1	top3	recall	recall	recall	recall	recall	recall	recall	recall
200	0.96	1	0.98	0.04	-	0.338	0.975	0.045	0.955	

Appendix C - CAPA evidence grade

association_evidence	temporal_evidence	adjusted_evidence	interfactual_evidence	causal_evidence	confidence_grade	0_4
post decline observed	pre-trend concern	supported	weak/inconclusive	uncertain	Inconclusive	1

Appendix D - Runtime scaling

events	feature_engineering_s	events_per_second	rss_delta_mb
100000	0.0625105	1.59973e+06	21.25
250000	0.137232	1.82173e+06	34.5195
504000	0.288532	1.74677e+06	98.8789

Part II — Formal statistical formulation and validation details

22. Formal notation and estimands

Let i index manufacturing events, s stations, m machines, o operators, v vehicle variants, l supplier lots, and t production days. For each event, the synthetic data-generating process produces an unobserved latent process state Z_i^* , a measured process value Z_i , a binary defect outcome Y_i , and a latent injected anomaly indicator A_i^* . The observed anomaly label A_i is imperfect and may differ from A_i^* because the simulation explicitly injects false negatives, false positives, and reporting delay.

The measurement model is written conceptually as

$$Z_i = Z_i^* + b_{m(i)} + \varepsilon_i,$$

where $b_{m(i)}$ is machine-specific measurement bias and ε_i includes repeatability noise and quantization. This distinction is important because a monitoring algorithm sees Z_i , not Z_i^* .

The primary anomaly-detection estimand is **not** classification accuracy. It is the probability that a latent anomaly is captured under a constrained investigation policy:

$$\Pr(A_i^* = 1 \mid S_i \geq q_{1-b}),$$

where S_i is an anomaly score and b is the review budget. In the main analysis $b = 0.0025$, so only 0.25% of evaluated events may be investigated.

The RCA estimand for a candidate binary factor X_i is an adjusted odds ratio,

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i + \gamma^T \mathbf{C}_i,$$

where $\mathbf{C}_i$ contains measured confounders such as shift and vehicle variant. The coefficient β_1 is interpreted as an **adjusted association** unless the causal identification assumptions are separately defended.

For ST07 CAPA evaluation, the target estimand is an average change in station-day defect probability relative to a counterfactual trajectory. The fixed-effects Difference-in-Differences specification is

$$Y_{st} = \alpha_s + \lambda_t + \beta \text{Treated}_s \times \text{Post}_t + \epsilon_{st}.$$

The coefficient β is not treated as identified merely because it is statistically different from zero. Parallel-trend plausibility, anticipation, common shocks, spillovers, and donor-pool quality are examined with an event study, synthetic control, placebo-in-time and placebo-in-space diagnostics.

For censored Weibull reliability analysis, the likelihood is

$$L(\beta, \eta) = \prod_{i:\delta_i=1} f(t_i | \beta, \eta) \prod_{i:\delta_i=0} S(t_i | \beta, \eta),$$

where $\delta_i = 1$ indicates a failure-terminated interval and $\delta_i = 0$ a right-censored interval. The final unfinished operating interval for each machine is retained explicitly instead of being discarded.

The CAPA financial model computes

$$NPV = -I_0 + \sum_{k=1}^T \frac{CF_k}{(1+r)^k},$$

but CF_k is random because the causal effect, annual exposure, unit loss, implementation cost and discount rate are uncertain. Therefore the decision object is the distribution of NPV, not a single deterministic point estimate.

23. Why temporal validation is the primary standard

Manufacturing histories violate the assumptions that make random splitting convenient. Process regimes evolve, tools age, maintenance policies change, supplier lots enter and leave, CAPA changes the state of the process, and production mix changes over time. A random split allows a model to learn from future regimes that would not be available at deployment time.

The final pipeline therefore uses a forward structure with three distinct periods: training, calibration and future test. The calibration interval is deliberately

separated from training so that fixed review-budget thresholds can be chosen without observing the future evaluation period. Model comparison uses the same windows and the same feature availability rules.

The random-split probe is retained only as a diagnostic experiment. It answers the question: *how much would performance appear to improve if we evaluated the same predictive problem using a conventional random split?* It does not define the official performance number.

This distinction also governs feature engineering. A variable may be perfectly legitimate for retrospective RCA but invalid for early warning. Defect flag, scrap flag, rework time and post-event containment outcomes are therefore excluded from the early-warning feature set even though they remain available to diagnostic and economic analysis.

24. Rare-event decision theory

When event prevalence is low, ROC-AUC can remain visually respectable even if the investigation queue is unusable. The operational metric is therefore constructed from the finite review budget.

For review budget b and threshold q_{1-b} estimated on calibration data,

$$\text{Precision@}b = \frac{TP_b}{TP_b + FP_b}, \quad \text{Recall@}b = \frac{TP_b}{TP_b + FN_b},$$

and

$$\text{Lift@}b = \frac{\text{Precision@}b}{\Pr(A^* = 1)}.$$

Lift is especially useful for an engineering audience because it compares model-guided review directly with random inspection at the same workload. It does not hide the absolute weakness of a detector: a high lift can coexist with low recall when the review budget is extremely narrow.

The final release therefore reports both. If 0.25% review captures only a small fraction of all latent anomalies, the correct conclusion is that the detector is a prioritization layer, not an autonomous quality gate.

25. Feature pruning and third-order complexity control

The initial feature set was intentionally broader than the final one. Candidate early-warning variables included normalized cycle time, process deviation, environmental temperature, tool-age exposure and changeover status. These variables are all industrially plausible, but plausibility is not sufficient reason to keep them in a predictive detector.

The six-seed ablation experiment applies a simple rule: a feature group is removed from the primary detector if its exclusion improves out-of-time PR-AUC consistently and its removal does not create a critical failure-mode blind spot that cannot be handled by another monitoring layer.

The result illustrates a central principle of this repository: **more data fields are not automatically more information**. Tool age belongs naturally in reliability and maintenance analysis. Cycle time belongs naturally in flow and downtime diagnostics. Ambient temperature can be valuable for specific thermal failure investigations. None of those facts requires the variables to be included in the quality early-warning model.

This is also why the release does not add deep autoencoders, transformers or large ensembles. A method is retained only when it solves a documented methodological risk or changes a decision.

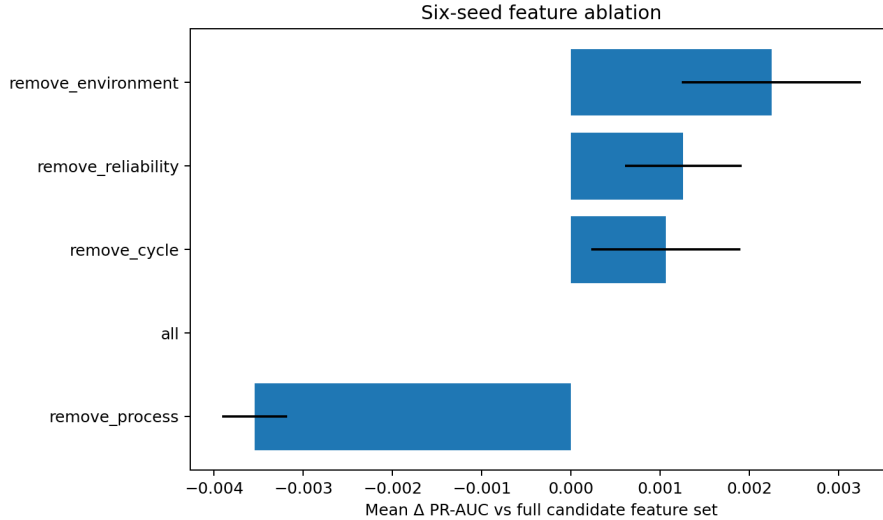


Figure 1: Feature ablation

26. Detector benchmarking under one decision rule

All benchmark detectors share the same temporal windows, feature availability, natural class prevalence and review budgets. This avoids a common benchmarking error in which each algorithm is allowed a favorable threshold or a different test sample.

The detector suite includes robust z-score, robust covariance/MCD, Isolation Forest, Local Outlier Factor, One-Class SVM and PCA residual monitoring. These methods represent different assumptions rather than a hierarchy of sophistication. Robust z-score assumes that marginal extremes are informative.

MCD models multivariate distance under a robust elliptical approximation. Isolation Forest exploits random partition depth. LOF is local-density based. One-Class SVM estimates a nonlinear boundary. PCA residual monitoring detects departures from a low-dimensional reconstruction.

The final detector is not chosen from ROC-AUC alone. At the narrow 0.25% engineering review budget, One-Class SVM provides the strongest lift in the current benchmark, while MCD and Isolation Forest provide similar global PR-AUC. This disagreement is itself useful: the model with the best global ranking is not necessarily the model with the best extreme-tail queue.

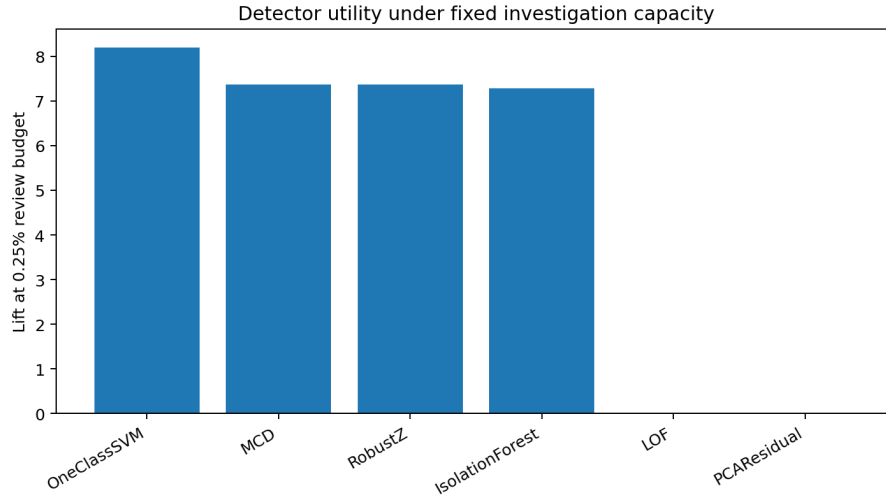


Figure 2: Detector lift

27. Failure taxonomy and capability matrix

The synthetic DGP deliberately includes mechanisms that differ in temporal shape and causal structure: step shifts, slow drifts, variance changes, rare spikes, temporary fixture issues, supplier-lot excursions, machine degradation, cyclical anomalies and interaction failures. A single aggregated PR-AUC cannot show whether a detector is useful for these different mechanisms.

For each failure mechanism and detector, the release reports event-level recall, within-window precision, median delay, P90 delay and whether the episode is detected at all. Missing detections remain missing; they are not imputed as zero-delay successes.

A major conclusion is that a process-value early-warning detector is not a universal manufacturing-health system. Some downtime-centric failures remain invisible in the process measurement and require dedicated downtime, telemetry or maintenance-state monitoring. This is why the final architecture retains

multiple specialized monitoring layers instead of combining everything into one opaque score.

28. SPC policy validation through run length

EWMA and CUSUM parameters are often chosen from standard references and then treated as if their behavior were guaranteed in every process. The release instead uses Monte Carlo run-length experiments as a policy comparison tool.

The in-control average run length, ARL_0 , approximates how long the chart runs before a false signal under the reference process. ARL_1 measures detection speed after a specified shift. The simulation compares in-control data, 0.5-sigma and 1-sigma step shifts, gradual drift and variance inflation.

These experiments are not claimed to reproduce exact plant ARLs because the synthetic event stream is not iid Gaussian in every regime. Their purpose is comparative: they show the relative detection burden and delay of the selected chart policies.

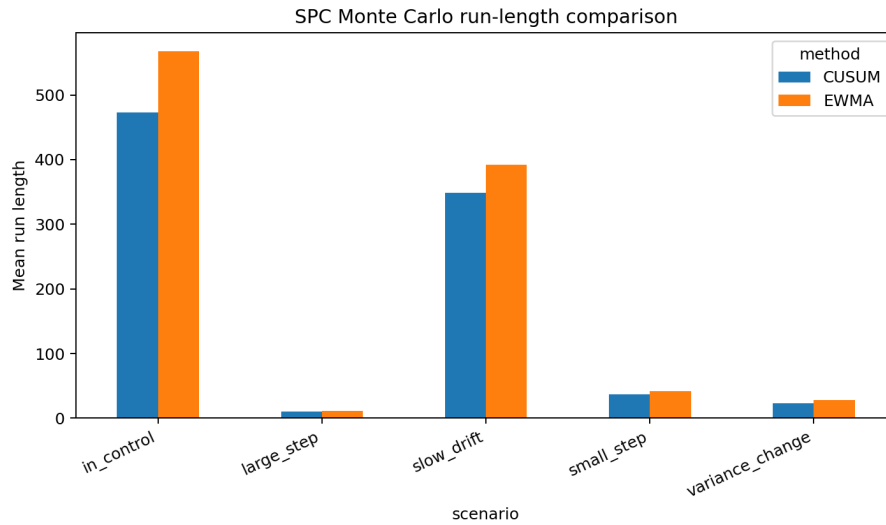


Figure 3: SPC run-length experiment

29. Root-cause inference as an evidence ladder

The RCA architecture intentionally prohibits a one-step transition from high anomaly score to root cause. Evidence progresses through levels:

1. anomaly association;
2. stratified association;
3. adjusted statistical association;

4. temporal precedence;
5. quasi-experimental evidence;
6. intervention verification.

A candidate can be useful at one level and unsupported at another. For example, a machine may have a materially elevated adjusted odds ratio and therefore deserve engineering investigation. That does not prove that a specific physical failure mechanism is the causal source of the defect. Physical confirmation would require inspection, calibration records, tool analysis or an intervention that changes the hypothesized mechanism.

The evidence graph stores these distinctions explicitly. It links anomaly, affected KPI, candidate cause, alternative explanations, statistical adjustment, CAPA and verification evidence. The graph is intended as a navigable audit trail rather than a causal discovery algorithm.

30. Firth logistic regression and sparse manufacturing evidence

Rare failures and highly unbalanced categories can cause complete or quasi-complete separation in ordinary logistic regression. In those cases, maximum-likelihood coefficients can diverge and standard errors become unstable.

The Firth comparator uses a Jeffreys-prior penalty to reduce small-sample bias and provide finite estimates under separation. It is not used because it is “more advanced”; it is used because it addresses a specific sparse-data failure mode. Agreement between ordinary logistic and Firth estimates provides a useful stability check. Divergence would be treated as evidence that standard asymptotic inference is unreliable.

The hierarchical comparator serves a different purpose. Shrinkage prevents small machine or supplier groups from dominating rankings solely because of extreme rates based on tiny denominators. These two methods therefore address different risks: Firth addresses likelihood bias/separation, while hierarchical shrinkage addresses noisy group-level ranking.

31. Simpson-type confounding experiment

The ST07 operator assignment is deliberately structured so that OP-07-01 is disproportionately exposed to the problematic M07B machine while receiving a modest conditional protective effect. This creates an environment in which aggregate operator rates can suggest a different story from an adjusted model.

The purpose is not to manufacture a textbook paradox perfectly in every individual stratum. The engineering lesson is that exposure assignment is not random. Operators, machines, shifts and variants are operationally coupled. Any RCA process that ranks people without accounting for machine assignment can create a false attribution.

The final adjusted operator odds ratio is below one and statistically inconclusive. The repository therefore classifies the raw operator story as falsified.

32. Known-truth root-cause recovery

Synthetic data provide one advantage that real plant data rarely provide: the true injected parameter is known. The release uses that advantage directly.

Across repeated datasets, the RCA experiment records whether the true machine factor is ranked first, whether it appears in the top three, its mean reciprocal rank, effect-estimation bias, RMSE and confidence-interval coverage. A separate null DGP contains no supported machine/operator cause. The pipeline is required to return *inconclusive* unless its statistical evidence crosses the predeclared support threshold.

This changes the meaning of the synthetic exercise. The question is no longer “can the model find something interesting?” It becomes “under what conditions does the analysis recover a known truth, and how often does it invent a cause when none exists?”

33. Multiple testing and evidence governance

Manufacturing RCA routinely screens many machines, operators, suppliers, lots and shifts. If each candidate is tested at 5% without correction, false discoveries accumulate rapidly.

The repository therefore uses Benjamini-Hochberg FDR as the primary multiple-testing policy when the goal is screening a family of potential causes, with Holm correction as a more conservative sensitivity check. Effect magnitude and uncertainty are reported with adjusted evidence. A small p-value without a material effect size does not automatically become an engineering priority.

Negative controls provide an additional falsification layer. The DGP includes variables designed to be irrelevant. If those variables appear repeatedly among top causes, the RCA pipeline is overfitting or failing to account for multiplicity.

34. Reliability: censored life versus repairable systems

The study distinguishes two reliability questions that are often mixed incorrectly.

A censored Weibull fit asks about the distribution of interval life while respecting operating intervals that have not yet ended in failure. Its final survival contribution must remain in the likelihood.

A repairable-system process asks how recurrent failures accumulate over operating time. Crow-AMSAA/NHPP behavior is therefore reported separately. A beta below one can be interpreted as improving failure intensity, a beta above one as deterioration, subject to model adequacy.

MTBF and MTTR remain useful descriptive summaries but are not treated as sufficient stochastic models.

35. CAPA identification and falsification

The ST07 CAPA result demonstrates why statistical significance is not identical to causal identification. The DiD coefficient is negative and its standard confidence interval excludes zero. If the analysis stopped there, the conclusion would look strong.

The event study changes that interpretation. Several pre-intervention coefficients are materially different from zero, indicating that the treated station was not following a clean parallel pre-trend relative to controls. Synthetic control provides an alternative counterfactual, but donor-pool size and pre-fit quality limit inferential strength. Placebo interventions assigned to other dates and other stations show that effects of comparable magnitude are not exceptionally rare.

For these reasons the final CAPA grade is Weak rather than Strong. The result is still useful operationally: it supports continued monitoring and may justify the action when economics and engineering mechanism are favorable. It does not justify claiming that the repository has proven a causal production effect.

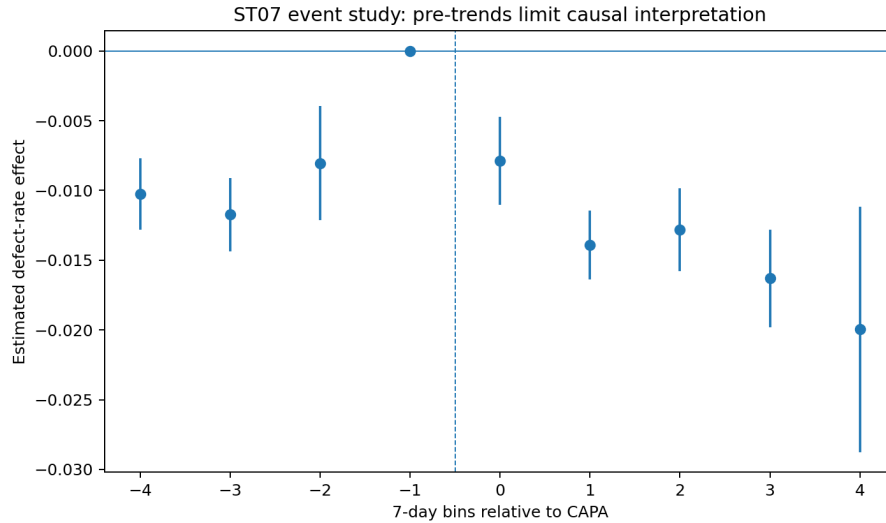


Figure 4: ST07 event study

36. CAPA heterogeneity

Treatment effects are not assumed to be homogeneous. Shift and vehicle-variant heterogeneity are estimated at station-day-subgroup level rather than raw event level, which reduces pseudo-replication. Machine-level differences within ST07 are reported only descriptively because they do not share the same cross-station counterfactual design.

This distinction is important. Subgroup tables are easy to generate and easy to overinterpret. The release only promotes heterogeneity results when the statistical unit and comparison design remain coherent.

37. Probabilistic economics and causal shrinkage

The financial model treats causal identification uncertainty as part of the business case. Because the CAPA evidence grade is weak, the DiD effect is deliberately shrunk toward zero before it is propagated to avoided defects. This is a conservative decision-model choice, not an estimate that emerges from a Bayesian causal model.

The remaining uncertainty is propagated through annual exposure, loss per defect, implementation cost, recurring cost and discount rate. The output therefore contains a positive median NPV and a negative downside tail. A manager can distinguish “expectedly attractive” from “guaranteed positive,” which are very different claims.

The 5% Conditional Value at Risk reports the mean outcome within the worst 5% of simulations. This makes the downside visible rather than hiding it behind the median.

38. Cost of Quality interpretation

The synthetic cost model follows the prevention-appraisal-internal-failure-external-failure structure. Prevention includes corrective engineering and recurring verification intended to stop recurrence. Appraisal includes investigation time and inspection burden. Internal failure captures modeled rework/scrap before customer escape. External failure is intentionally excluded from the base case where the synthetic DGP does not directly represent warranty or customer-return behavior.

That exclusion is important. Adding a large assumed warranty cost would make almost any CAPA look attractive. The final release prefers an incomplete but traceable economic model over a larger benefit number with weak provenance.

39. Human-in-the-loop workload

Every review budget is translated into analyst workload. For each detector and budget, the repository reports events investigated, true anomalies captured,

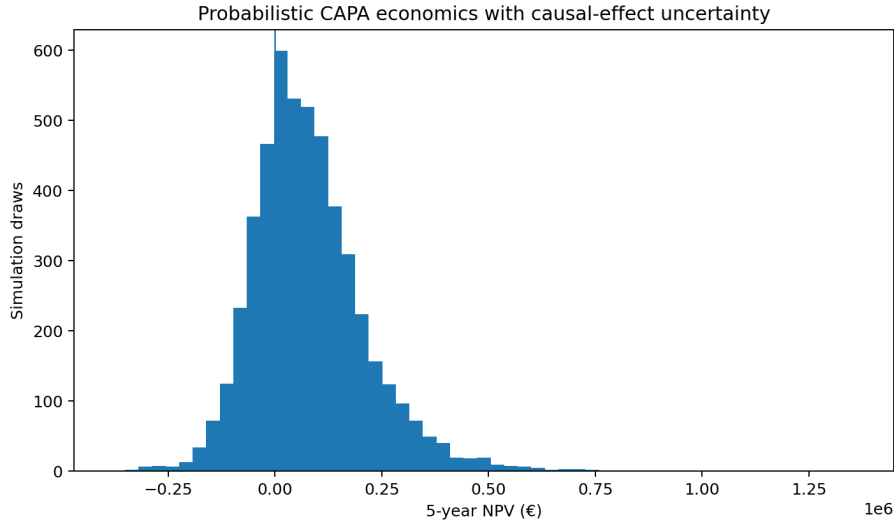


Figure 5: CAPA NPV distribution

false investigations, analyst hours and estimated investigation cost. This prevents a model from winning merely because it generates more alerts.

The human-in-the-loop framing also changes the role of anomaly detection. The model is a queue-prioritization tool. Engineers still need contextual evidence, maintenance history, supplier genealogy and physical verification before assigning root cause or initiating CAPA.

40. Staged pipeline, caching and resume semantics

A research-grade analytics repository must survive interrupted runs. Each stage therefore computes a signature from its configuration and upstream file hashes. The stage output manifest records the signature, hashes of generated artifacts, runtime and peak RSS. On rerun, an unchanged signature with matching output hashes causes the stage to be reused.

This mechanism is intentionally simple. It is not a replacement for an enterprise workflow orchestrator, but it makes local reproducibility explicit and testable. The full pipeline can be resumed after a failed stage without regenerating 504,000 events or refitting unrelated models.

41. Verification, validation and uncertainty quantification

The credibility case separates four ideas:

Code verification: do functions and stage contracts behave as intended?

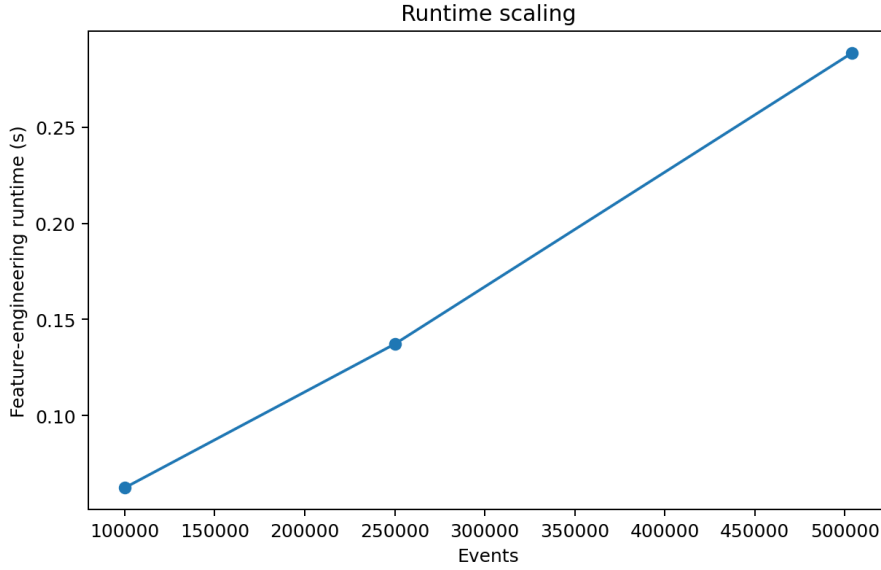


Figure 6: Runtime scaling

Statistical validation: do estimators recover known synthetic parameters with acceptable bias and coverage?

Algorithm validation: do detector rankings remain useful under future-regime and failure-type changes?

External validation: does the architecture describe a real manufacturing system accurately?

The first three are addressed partially within this repository. The fourth is explicitly absent. That boundary prevents synthetic validation from being presented as plant validation.

42. Runtime scalability

The largest source table contains 504,000 production events plus 300,000 telemetry observations. Feature-engineering scaling is measured at 100k, 250k and 504k rows. Stage-level manifests additionally record elapsed time and peak RSS.

The main engineering conclusion is that memory stability matters more than raw arithmetic speed. The earlier monolithic design was fragile because several large DataFrames and repeated model fits coexisted in one process. Staging removes that failure mode and makes expensive robustness experiments separable from the full-resolution production benchmark.

43. Final methodological limitations

Several limitations remain material.

First, the DGP is still authored. Even an adversarial synthetic generator cannot reproduce the unknown dependence structure of a real automotive plant.

Second, measurement-system behavior is simplified. A real MSA would require repeated physical parts, operators, gauges and documented reference standards.

Third, supplier and operator assignment are stylized. Real routing and labor allocation may introduce stronger selection mechanisms.

Fourth, the one-treated-station CAPA design has limited inferential power. A small donor pool constrains placebo resolution, and common shocks may not be fully removed.

Fifth, financial values are synthetic design assumptions. The economic model demonstrates uncertainty propagation, not actual financial return.

Sixth, the early-warning quality detector intentionally omits variables that help other operational tasks. It is not a line-wide condition-monitoring model.

Seventh, the null Monte Carlo experiment evaluates a particular class of DGPs. A low false-cause rate under those simulations does not guarantee the same rate under real-world unmeasured confounding.

44. External deployment requirements

A credible industrial pilot would require at minimum:

- MES event logs with stable event and routing identifiers;
- QMS defect codes and inspection timestamps;
- CMMS failures, work orders and maintenance completion times;
- supplier and lot genealogy;
- operator/crew assignment history;
- actual gauge and calibration records;
- process parameter traces with engineering units;
- CAPA effective dates and change-control documentation;
- finance-approved cost rates and warranty/escape data;
- versioned process standards and equipment configurations.

Point-in-time availability must be documented for every predictive feature. Without that provenance, temporal leakage can re-enter the system even when the model code itself is correct.

45. Final classification

The release is best classified as a **research-grade synthetic manufacturing analytics study with PhD-level methodological sophistication in selected areas**. It is stronger than a conventional portfolio project because it

validates parameter recovery, false discovery, censoring treatment, fixed-budget utility, causal falsification and probabilistic economics. It is not an actual publishable research contribution because it does not establish methodological novelty or validate the system on independent empirical manufacturing data.

The bottleneck is therefore external validity, not algorithm count.

Appendices

Appendix E - Release audit summary

Critical weaknesses (3)

- **Look-ahead percentile scoring** - risk: temporal leakage; correction: Replaced with train/calibration/test thresholds.
- **Post-outcome consequence fields in early warning** - risk: feature leakage; correction: Separated early-warning from post-event triage.
- **Random split as optimistic comparator** - risk: temporal validation; correction: Chronological split is primary. ## High weaknesses (22)
- **Rare-event ROC emphasis** - risk: metric bias; correction: PR-AUC and fixed-budget lift are primary.
- **Detector chosen by aggregate AUC** - risk: decision mismatch; correction: Select by review-budget utility and regime coverage.
- **Environmental feature instability** - risk: distribution shift; correction: Pruned from early-warning detector.
- **Tool-age feature instability** - risk: model variance; correction: Pruned from early-warning detector.
- **Cycle-time feature instability** - risk: model variance; correction: Pruned from quality early-warning detector; retained in diagnostics.
- **Single synthetic realization** - risk: validation weakness; correction: Known-truth Monte Carlo recovery experiment.
- **No null RCA scenario** - risk: false discovery; correction: Null simulations can return inconclusive.
- **Negative controls absent** - risk: spurious discovery; correction: Two irrelevant features retained and tested.
- **Raw Pareto causal interpretation** - risk: confounding; correction: Adjusted compact logistic + evidence hierarchy.
- **Operator-machine assignment confounding** - risk: misattribution; correction: Deliberate Simpson-type scenario and adjustment.

- **Rare-event separation risk** - risk: bias; correction: Firth comparator added.
- **Small supplier samples** - risk: ranking noise; correction: Hierarchical/Beta-Binomial logic retained.
- **Failure-only Weibull** - risk: censoring bias; correction: Right-censored Weibull likelihood.
- **Repairable system treated as lifetime data** - risk: model mismatch; correction: Crow-AMSAA reported separately.
- **CAPA before/after claim** - risk: causal overclaim; correction: DiD + event study + synthetic control + placebos.
- **CAPA subgroup pseudo-replication** - risk: false precision; correction: station-day-subgroup units.
- **Economic benefit from raw before/after** - risk: double attribution; correction: Causal-effect shrinkage and uncertainty propagated.
- **Deterministic NPV** - risk: decision risk; correction: Monte Carlo NPV/payback/CVaR.
- **Monolithic pipeline** - risk: reproducibility; correction: 10 independent cached stages.
- **No resume capability** - risk: operational fragility; correction: Hash manifests and output validation.
- **Huge DataFrames retained** - risk: memory risk; correction: Stage isolation and explicit release.
- **Parquet dependency not available in runtime** - risk: environment risk; correction: Verified gzip-pickle fallback; Parquet recommended for deployment. ## Low weaknesses (2)
- **Too many redundant models** - risk: complexity; correction: Deep learning/ensemble novelty pruned.
- **Scientific novelty overclaim** - risk: credibility; correction: Classified as research-grade synthetic study, not publishable contribution. ## Medium weaknesses (18)
- **Label assumed perfect** - risk: label risk; correction: Observed label noise and latent truth separated.
- **No detector false-alert workload** - risk: actionability; correction: Analyst hours and investigation cost.
- **No cost of detection delay** - risk: business blind spot; correction: Failure-type delay cost table.

- **No failure taxonomy** - risk: aggregated masking; correction: 14 mechanisms with explicit type.
- **No method-capability matrix** - risk: method selection risk; correction: Failure x detector metrics.
- **SPC ARL not benchmarked** - risk: policy uncertainty; correction: Monte Carlo ARL0/ARL1.
- **Autocorrelation ignored** - risk: SPC false alarms; correction: Residual-based approach documented when diagnostics require it.
- **Multivariate SPC could be decorative** - risk: complexity risk; correction: Pruned from final narrative unless covariance adds decision value.
- **Change-point methods treated as universal** - risk: false confidence; correction: Negative detection results preserved.
- **Multiple testing ignored** - risk: false discovery; correction: BH-FDR/Holm policy.
- **Effect size omitted** - risk: management ambiguity; correction: OR/RR/risk-difference governance.
- **Claim classes mixed** - risk: communication risk; correction: Claim governance matrix.
- **FMEA detached from evidence** - risk: quality-system gap; correction: Failure evidence linkage/control plan.
- **No reaction plan** - risk: deployment gap; correction: 5 synthetic control/reaction plans.
- **No 8D-style case** - risk: operational translation gap; correction: ST07 formal case study.
- **No machine-readable data contracts** - risk: integration risk; correction: Contracts + tests for five datasets.
- **No runtime scaling evidence** - risk: deployment uncertainty; correction: 100k/250k/504k benchmark.
- **No config provenance** - risk: reproducibility gap; correction: Config/hash stage manifests.

Appendix F - Claim governance

- **Supported: Quality early-warning adds value at a 0.25% review budget** Evidence: OneClassSVM lift 8.20x; precision 13.01%. Falsification: adversarial/failure-regime matrix.

- **Supported: Random split materially overstates temporal performance** Evidence: PR-AUC 0.0527 vs 0.0322. Falsification: random-vs-time comparison.
- **Falsified: Environmental/tool-age/cycle features belong in the primary early-warning model** Evidence: Mean ablation deltas: {'all': 0.0, 'remove_cycle': 0.0011, 'remove_environment': 0.0022, 'remove_process': -0.0035, 'remove_reliability': 0.0013}. Falsification: remove-one-group tests.
- **Falsified: Raw operator ranking proves operator responsibility** Evidence: Aggregate/adjusted direction changes in the ST07 confounding case. Falsification: Simpson/confounding experiment.
- **Partially supported: M07B is a supported ST07 RCA candidate** Evidence: Standard OR 2.58; Firth OR 2.49. Falsification: confounder adjustment.
- **Inconclusive: ST07 CAPA has a clean causal effect** Evidence: DiD -0.0025; time-placebo $p=0.600$; space-placebo $p=0.364$. Falsification: pre-trend + donor falsification.
- **Partially supported: ST07 CAPA is economically attractive with certainty** Evidence: $P(NPV > 0) = 0.728$; $P5 = \text{€-}97,369$; $CVaR5 = \text{€-}150,116$. Falsification: negative tail retained.
- **Supported: The RCA pipeline can refuse unsupported causes** Evidence: Null inconclusive rate=0.955. Falsification: false-RCA declaration test.

Appendix G - Failure capability highlights

- cyclic_vibration / IsolationForest: recall 0.00%, not detected.
- cyclic_vibration / MCD: recall 0.51%, median delay 69.0 d, P90 69.0 d.
- cyclic_vibration / RobustZ: recall 0.51%, median delay 69.0 d, P90 69.0 d.
- pneumatic_leak / IsolationForest: recall 0.00%, not detected.
- pneumatic_leak / MCD: recall 0.00%, not detected.
- pneumatic_leak / RobustZ: recall 0.00%, not detected.
- sensor_offset_drift / IsolationForest: recall 0.27%, median delay 62.0 d, P90 62.0 d.
- sensor_offset_drift / MCD: recall 0.27%, median delay 62.0 d, P90 62.0 d.
- sensor_offset_drift / RobustZ: recall 0.27%, median delay 62.0 d, P90 62.0 d.
- short_intense_failure / IsolationForest: recall 11.11%, median delay 1.5 d, P90 2.0 d.
- short_intense_failure / MCD: recall 13.89%, median delay 2.0 d, P90 2.0 d.
- short_intense_failure / RobustZ: recall 13.89%, median delay 2.0 d, P90 2.0 d.

- `supplier_wear_interaction` / IsolationForest: recall 0.00%, not detected.
- `supplier_wear_interaction` / MCD: recall 0.00%, not detected.
- `supplier_wear_interaction` / RobustZ: recall 0.00%, not detected.
- `temporary_fixture_issue` / IsolationForest: recall 0.00%, not detected.
- `temporary_fixture_issue` / MCD: recall 0.00%, not detected.
- `temporary_fixture_issue` / RobustZ: recall 0.00%, not detected.
- `tool_wear` / IsolationForest: recall 3.62%, median delay 51.0 d, P90 56.4 d.
- `tool_wear` / MCD: recall 3.62%, median delay 51.0 d, P90 56.4 d.
- `tool_wear` / RobustZ: recall 3.62%, median delay 51.0 d, P90 56.4 d.
- `variance_shift` / IsolationForest: recall 4.95%, median delay 6.0 d, P90 24.4 d.
- `variance_shift` / MCD: recall 4.95%, median delay 6.0 d, P90 24.4 d.
- `variance_shift` / RobustZ: recall 5.49%, median delay 8.0 d, P90 23.7 d.
- `weak_quality_drift` / IsolationForest: recall 0.32%, median delay 51.0 d, P90 51.0 d.
- `weak_quality_drift` / MCD: recall 0.32%, median delay 51.0 d, P90 51.0 d.
- `weak_quality_drift` / RobustZ: recall 0.32%, median delay 51.0 d, P90 51.0 d.

Appendix H - Control and reaction plan summary

- **Calibration drift:** monitor `process_z` + EWMA/CUSUM; trigger Phase-II EWMA/CUSUM signal; owner Quality Engineering; containment hold affected station output; verify gauge; verification post-action SPC stability + MSA.
- **Supplier lot excursion:** monitor lot-linked defect rate / incoming inspection; trigger posterior lot risk + traceability excursion; owner Supplier Quality; containment segregate lot; increase incoming sampling; verification lot exit and post-containment defect normalization.
- **Pneumatic microstops:** monitor downtime event rate + pressure telemetry; trigger rate exceeds station reference; owner Maintenance; containment safe-stop and restore pressure; verification microstop rate and MTTR.
- **Tool wear:** monitor cycle/process tail + maintenance exposure; trigger wear trend or life threshold; owner Maintenance Engineering; containment planned tool swap if threshold exceeded; verification post-change cycle/quality distribution.
- **Variance shift:** monitor process dispersion / robust sigma; trigger dispersion > reference band; owner Quality Engineering; containment contain product since last verified stable point; verification variance returns within control reference.

Appendix I - Reproducibility

```
python -m src.staged_pipeline_v3 --root . --config configs/research_grade.yaml --stage all  
pytest -q
```

Full machine-readable appendices remain in `outputs/v3/*.csv` and the complete Markdown thesis.